Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



# Team Details

**Inventory Rescue**

a. Team name: Fantastic Four
b. Team leader name: Ashish Kumar Singh
c. Problem Statement: Growing food & nutrition brands lose lakhs a month to stockouts and expired stock because supplier delays and demand spikes arrive as unstructured messages, and limited cash forces hard buy / expedite / discount trade-offs.

# Brief about the idea

*"With ₹10 lakh available this month, what should we buy, expedite, discount or defer, and when?"*

## The problem

- Mid-size and large D2C / FMCG brands (₹50 crore+ a year) still plan rescues on spreadsheets, email and WhatsApp.
- Supplier delays, influencer-driven spikes and expiring lots arrive as unstructured messages.
- Cash is tight, so hero SKUs stock out while slow movers expire and get written off.

## Our solution

- Generative AI reads the messages (English, Hindi, Hinglish), turns text-only news into forecast adjustments and answers "why / what if" questions with planner tools.
- A deterministic simulator and optimiser decide quantities and money; every LLM output is validated.
- The operator confirms signals, picks when to spend and exports POs.

## The outcome

- One ranked action list within budget, plus four ways to time the spend (commit now, staggered, reserve, lean).
- Demo brand (₹4.3 crore revenue a month): **₹3.8 lakh / month** lower simulated loss than a static reorder rule and **₹1.8 lakh** than a tuned one, better in **120 of 120** simulated futures.
- Keeping 30% in reserve for weekly reviews: a further **₹0.9–1.1 lakh** in 9–10 of 10 futures.

Target users: starting with supply-chain and operations teams of mid-size to large FMCG / D2C brands across JAPAC; the engine is product-agnostic. All values are simulated on seeded data and will be validated in pilots.

# Opportunities

## How is it different from existing ideas?

- **Inventory apps / ERPs** track stock but use static reorder points and stated lead times, and ignore supplier or campaign messages.
- **Enterprise demand planners** forecast well but take months to roll out and still ignore the inbox; the rescue decisions happen in spreadsheets.
- **Generic AI chatbots** give advice without arithmetic and can invent numbers.
- **Ours** pairs LLM understanding with budget-aware optimisation and a built-in proof of value.

## How will it solve the problem?

- Turns messages into typed events with verbatim evidence; a human confirms.
- Hybrid forecasts: a statistical base, with the LLM tuning event effects that only exist as text.
- Supplier delay history gives early / typical / late arrival cases.
- Picks the best action mix under one cash budget, tests it on 120 simulated futures and shows when to spend.

## USP of the proposed solution

- **One cash budget** across buy, expedite, markdown and defer, and over time (staggered buys).
- **Trust by design:** the LLM proposes, code validates; every number in an answer is checked.
- **Measured, not claimed:** extraction gold set, forecast backtest, Monte Carlo, rule baselines.
- **Built on Google Cloud:** Gemini on Vertex AI, Cloud Run, Firebase Hosting, Firestore, Cloud SQL.

**Why now:** LLMs can finally read messy supplier and campaign messages well enough, when wrapped in validation, to feed real planning maths, and run it on pay-per-use cloud for a fraction of what it saves.

# List of features offered by the solution

### 1 Read a message

Paste emails or WhatsApp notes (also Hindi/Hinglish); Gemini finds delays, spikes, drops and expiry with verbatim quotes. You say yes or no.

### 2 Hybrid forecasts

Statistical base (Holt-Winters / Croston). The LLM tunes event effects within bounds, including news that exists only as text.

### 3 Ask a question

Ask “why” or “what if” in plain language. Gemini calls 11 read-only planner tools; every number is checked.

### 4 Grounded explanations

Plan narrative and what-if parsing that may only use numbers and IDs from the plan; else a labelled template.

### 5 Supply & service level

Supplier delay cases (early / typical / late); hero SKUs carry a goodwill penalty per unit short.

### 6 Budgeted planner

Buy, expedite, markdown (with cannibalisation) or defer; an exact knapsack picks the best set under the cash budget.

### 7 Plan options

Commit now, staggered buys, keep a reserve or lean, compared with weekly re-planning in calm and shocked futures.

### 8 Robustness & proof

120 Monte Carlo futures vs static and tuned reorder rules; budget frontier; forecast backtest.

### 9 Simple & Advanced

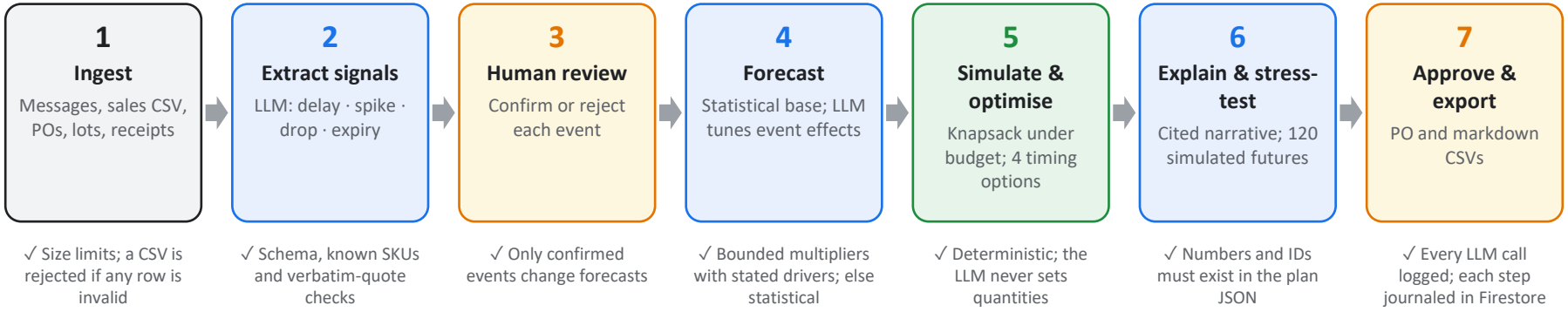
A 3D home page and three big buttons for anyone; the Advanced view adds charts, alerts, overrides and plan diffs.

### 10 Approve & journal

Approve to export PO and markdown CSVs; every message, decision, plan and question is journaled in Firestore.

Blue = generative AI · Green = deterministic engine · Orange = operator workflow

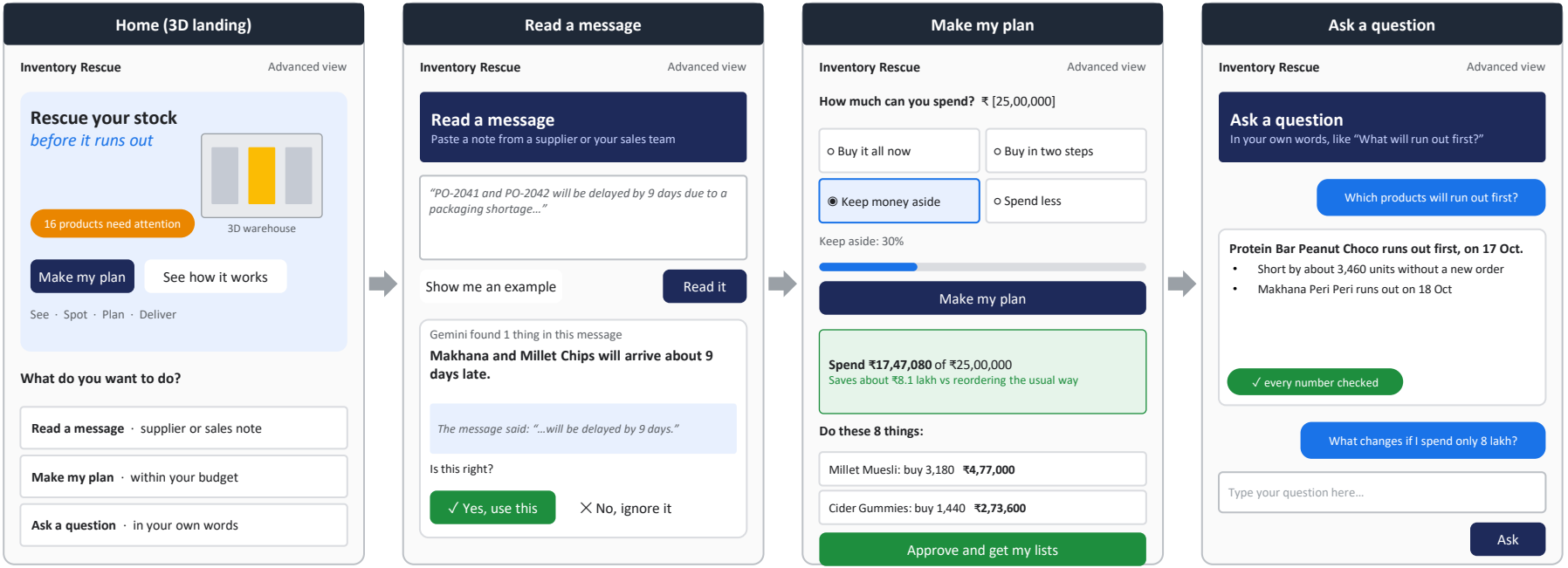
# Process flow diagram




↻ **Weekly reviews (days 0 / 7 / 14 / 21):** realised sales re-forecast → remaining and scheduled cash re-planned → staggered tranches re-confirmed, resized or redirected

**Ask the assistant:** “What if SUP-04 slips another week?” → the model calls read-only planner and simulator tools → answer with a tool trace; every number checked against tool results

# Wireframes / mock diagrams of the proposed solution




**1.** See what is at risk

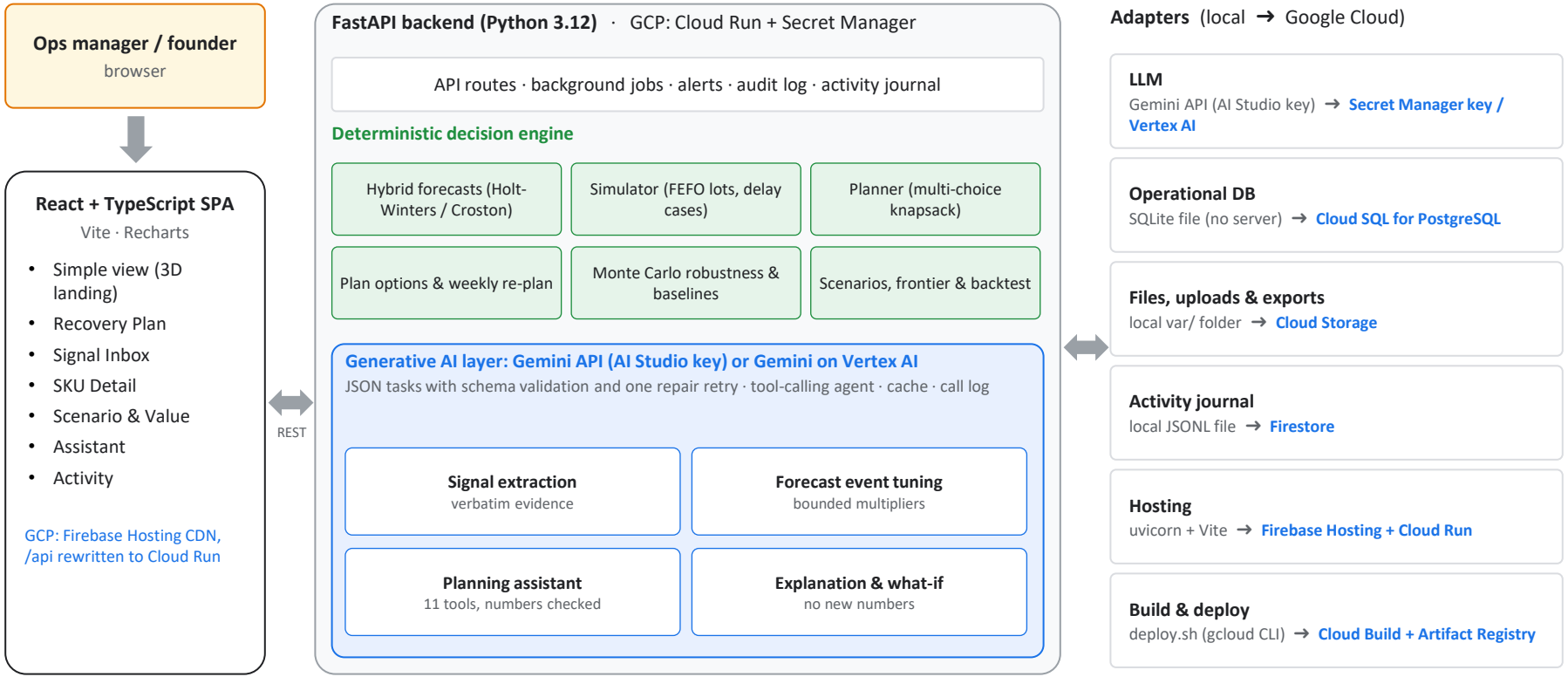
**2.** Confirm what Gemini read

**3.** Plan within budget, approve

**4.** Ask, with checked numbers

Simple view (default) shown. The Advanced view (Recovery Plan, Signal Inbox, SKU Detail, Scenario & Value, Assistant, Activity journal) is one click away.

# Architecture diagram of the proposed solution



# Technologies to be used in the solution

| Layer | Prototype (built and running) | Google Cloud (production) |
|---|---|---|
| **Frontend** | React 19, TypeScript, Vite, Recharts, three.js | Firebase Hosting (CDN), /api rewritten to Cloud Run |
| **Backend API** | Python 3.12, FastAPI, Pydantic v2, SQLAlchemy 2, Uvicorn | Cloud Run, Secret Manager |
| **Generative AI** | google-genai, gemini-3.5-flash-lite via the Gemini API or Vertex AI; validated JSON tasks; tool-calling assistant | Gemini API (key in Secret Manager) or Gemini on Vertex AI |
| **Decision engine** | Holt-Winters and Croston/SBA in NumPy; stochastic simulator; exact knapsack | Same code in Cloud Run (CPU always allocated) |
| **Data & files** | SQLite (no database server); local folder for uploads, exports, forecast cache | Cloud SQL for PostgreSQL, Cloud Storage, Firestore (activity journal) |
| **Evaluation** | Extraction gold set, rolling backtest, Monte Carlo, rule baselines | Same jobs; CSV exports ready for BigQuery |
| **Quality & delivery** | pytest (162 tests), oxlint, strict TypeScript, LLM call log | Cloud Build → Artifact Registry → Cloud Run |

# Estimated implementation cost

| Component (one brand, production sizing) | US$ / month |
|---|---|
| Cloud Run: API + UI, 2 vCPU / 2 GiB, one always-on instance | 90 – 105 |
| Cloud SQL for PostgreSQL 16: db-g1-small, 10 GB | 25 – 40 |
| Gemini 3.5 Flash-Lite (Vertex AI): ~10M input + 1M output tokens | 5 – 10 |
| Cloud Storage, Secret Manager, Artifact Registry, Cloud Build | < 3 |
| Firestore journal + Firebase Hosting (free tier at this volume) | < 1 |
| **Total running cost, single tenant** | **≈ 120 – 160** |
| **Lean pilot: 1 vCPU / 1 GiB, background jobs on Cloud Tasks, scale to zero** | **≈ 35 – 55** |

Indicative list prices (asia-south1); confirm with the Google Cloud Pricing Calculator. Excludes team effort. Cost does not grow with revenue: a 300-SKU catalogue adds roughly US$ 50–90 of Gemini calls a month on the same servers. Smaller brands can share one multi-tenant deployment.

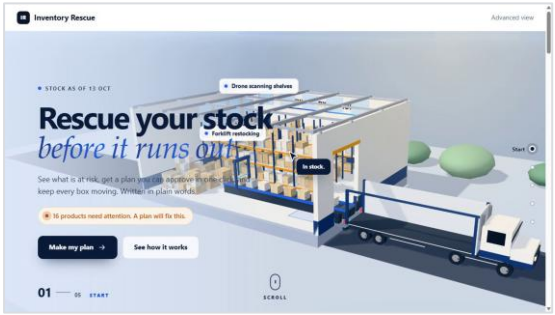
## Usage assumptions

- Daily re-forecast of 30 SKUs (≈900 forecast calls a month).
- ≈300 plans, 600 messages and 1,000 assistant questions a month.
- Measured LLM outputs are small: 0.2–2.3k characters per call.

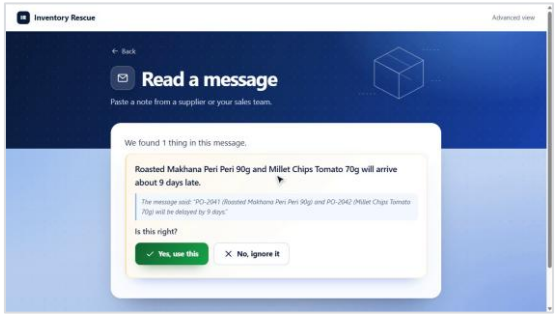
## Cost vs value

- Dedicated deployment ≈ ₹10.5k–14k a month (at ₹88 / US$).
- Simulated loss avoided for the demo brand (₹4.3 crore revenue a month): ₹1.6–4.0 lakh a month.
- ≈ 11–38× the running cost. At a twentieth of this size (≈ ₹2.6 crore a year) a dedicated instance only breaks even, so small brands should share one.

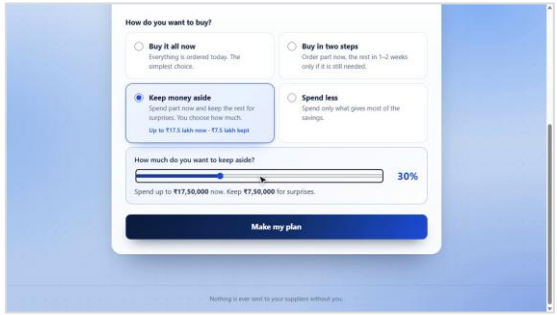
# Snapshots of the prototype



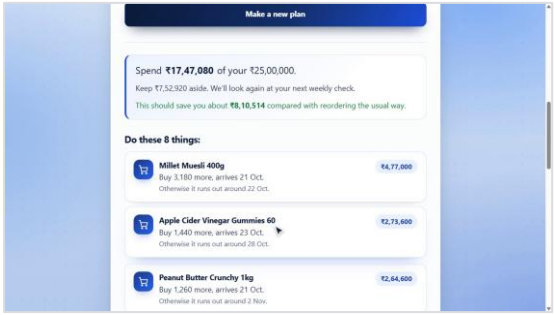

1. Opens on a 3D warehouse: what is at risk, in one line.



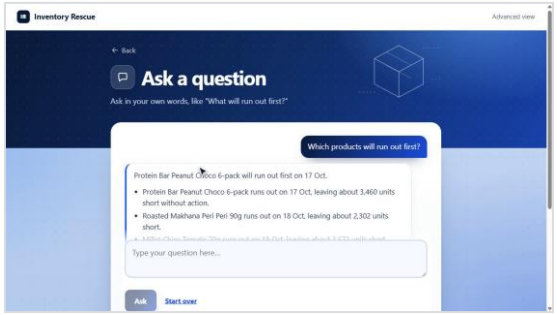

2. Read a message: Gemini quotes the line it used; you say yes or no.



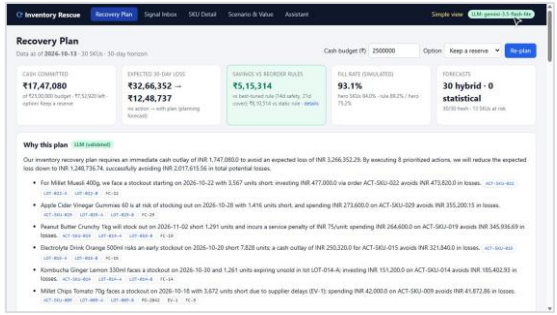

3. Type a budget; choose how to buy and how much to keep aside.




4. A plain to-do list: what to buy, by when, and the saving.



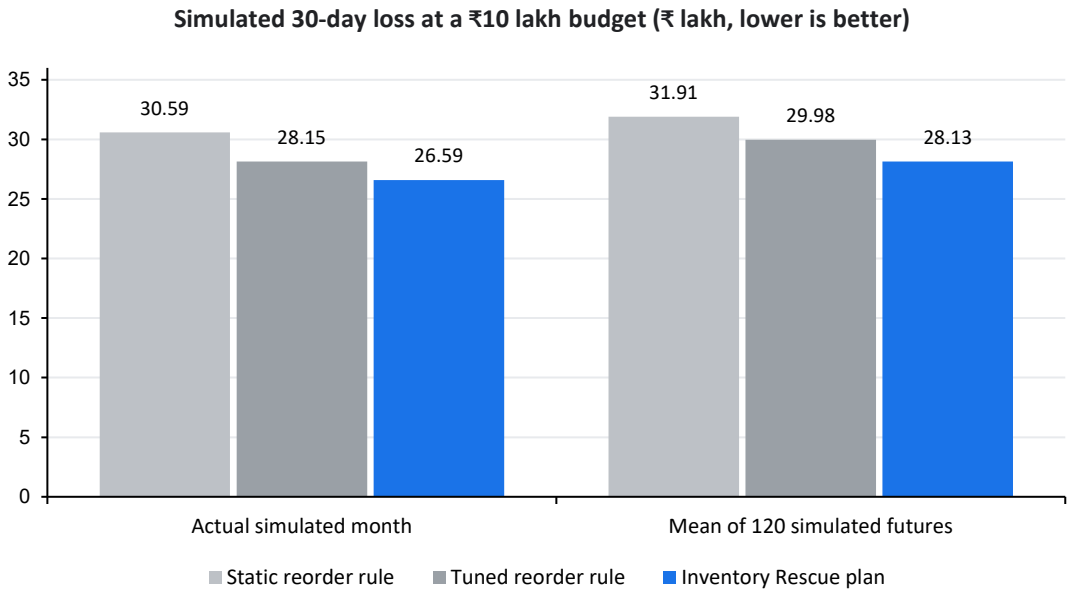

5. Ask in your own words; Gemini answers from checked numbers.




6. Advanced view: every number in the plan, with its evidence.

The Simple view is the default: a short 3D landing page, then three big buttons. Charts, robustness tests and scenarios stay one click away in the Advanced view.

# Prototype performance report / benchmarking




Every policy starts from the same stock and budget and faces the same simulated demand and supplier delays. Fill rate 89.7% vs 88.6–88.8% for the rules; hero-SKU fill 80.3% vs 75.2%.

**Engineering:** 162 automated tests pass · 30 / 30 seed forecasts passed LLM validation · every LLM call logged with its status.

**₹3.8 lakh / ₹1.8 lakh a month lower loss**
Mean vs static / tuned reorder rule; better in 120 of 120 futures

**30% reserve: ₹0.9–1.1 lakh further gain**
Re-planned at weekly reviews; better in 10 / 10 calm and 9 / 10 shocked futures

**Extraction F1 0.97 vs 0.65**
LLM + validators vs keyword rules, 40 labelled messages; plans from LLM signals keep 100% of the gold-signal gain

**Forecast WAPE 21.5% vs 23.0%**
Hybrid vs statistical, 1,050 points (22.2% vs 23.9% on 16 text-only series)

**Assistant: 6 / 6 answers number-checked**
5 on the first try, 1 after one repair; 3–10 s each (gemini-3.5-flash-lite)

# Gemini in the loop: what gemini-3.5-flash-lite adds

| Task | With Gemini | Without the LLM | Guardrail |
|---|---|---|---|
| **Read inbox messages into signals (40 labelled: email, WhatsApp, Hindi)** | **F1 0.97; 95% of messages fully correct; 0 invented events** | Keyword rules: F1 0.65; 57.5% correct | Evidence quote must appear verbatim; SKU, date and quantity checked |
| **Use those signals in a ₹10 lakh plan** | **100% of the perfect-information gain (actual month and 40 worlds)** | Hand-labelled signals = 100% | A person confirms each signal before it is used |
| **Tune event effects in the forecast (25 SKUs, 1,050 points)** | **WAPE 21.5%; P10–P90 coverage 76%** | Statistical 23.0% / 75%; seasonal naive 24.1% | 75/75 outputs passed validation; else statistical |
| **Events known only from text (16 series)** | **WAPE 22.2%** | Statistical 23.9% | Same validators |
| **Planning assistant with tool calls (6 questions)** | **6/6 answers number-checked (5 first try); median 4.0 s** | Labelled summary of the tool results | Every number must come from a tool result |

**Why gemini-3.5-flash-lite:** fast (median assistant answer 4.0 s, tool calls included), every output passes the validators, and it costs about US$ 5–10 a month at pilot volume. The live app calls it on Vertex AI; every call is logged with its status and a failed check falls back to a labelled non-LLM result.

**Next:** these runs used the AI Studio free tier (20 requests a day, about 130 per run), so larger models such as gemini-3.8-flash are not yet measured; with Vertex AI quota we will compare them and route each task to its best model. Small sets: 40 messages, 75 forecasts, 6 questions, so gaps under about 0.5 pp WAPE are noise.

# Additional details / future development

## Product

- Multi-warehouse and channel allocation (marketplace vs D2C).
- Email / WhatsApp connectors and auto-drafted supplier follow-ups.
- SAP, Unicommerce and Shopify connectors for stock and POs.
- Plan for signal uncertainty: one misread day can move ₹1 lakh.
- Learn event multipliers from realised outcomes.

## Platform on Google Cloud

- Live today on Cloud Run, Cloud SQL, Firestore and Firebase Hosting; next: HA and per-tenant data.
- Benchmark larger Gemini models on a paid key; route each task to its best model.
- Background jobs on Cloud Tasks; multi-tenant auth via Identity Platform.
- BigQuery ML forecasts as a second benchmark; Looker dashboards.

## Validation & go-to-market

- Pilot with 2–3 mid-size FMCG / D2C brands on their real data.
- Measure realised savings against the simulated scorecard.
- Grow the gold set with real, operator-labelled messages.
- Pricing: subscription per SKU band, well below the savings.
- **Beyond FMCG:** pharma and cosmetics as-is; electronics, fashion and spare parts with expiry switched off.

**Responsible AI:** the LLM never sets quantities or money; every output is validated, labelled and logged, and signals need human confirmation.

# Links

**GitHub public repository**
https://github.com/Xhahat/inventory-rescue

**Demo video (3.5 minutes)**
https://drive.google.com/file/d/1hqANKrCaGkMQFxnXu5LMdp8DBv_WMhHt/view

**Final product link**
https://inventory-rescue-511705.web.app · Firebase Hosting + Cloud Run (asia-south1)

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# Thank you

